

An AI-Driven Personalized Sinhala Teaching Assistant for Secondary Education Using Curriculum Aligned Large Language Models

Abstract—Sri Lanka faces a shortage of qualified teachers, with rural regions being disproportionately affected, resulting in unequal learning opportunities for students preparing for the General Certificate of Education Ordinary Level (GCE O/L) and Advanced Level (GCE A/L) examinations. Existing digital learning platforms primarily provide static, prerecorded content and lack adaptive personalization, while general-purpose Large Language Models (LLMs) are not aligned with the national curriculum. This paper presents an Artificial Intelligence (AI)-driven Sinhala teaching assistant that integrates four components: (i) a curriculum-aligned Sinhala LLM (SinLlama-7B) for educational content generation, (ii) a Hybrid Bayesian Knowledge Tracing (Hybrid-BKT) framework combining Personalized Clustered Bayesian Knowledge Tracing with Long Short-Term Memory (LSTM)-based temporal modeling for learner mastery estimation, (iii) a Retrieval-Augmented Generation (RAG) pipeline supported by a ChromaDB indexed national curriculum knowledge base, and (iv) a LangGraph-orchestrated multi-agent architecture for coordinated instructional support. SinLlama-7B outperforms Qwen-14B, Llama-3.1-8B, DeepSeek-14B, and Mistral-7B in curriculum alignment and semantic fidelity. The Hybrid-BKT framework continuously estimates learner knowledge states, enabling adaptive content delivery tailored to individual proficiency levels. Experimental results demonstrate the feasibility and effectiveness of integrating curriculum-aligned language models, knowledge tracing, and retrieval-augmented generation for personalized secondary education in Sri Lanka.

Keywords—*Bayesian knowledge tracing, Intelligent tutoring systems, Personalized Learning Systems, Retrieval-augmented generation (RAG), Sinhala natural language processing (NLP)*

I. INTRODUCTION

Education is widely recognized as a cornerstone of human development, yet students preparing for the General Certificate of Education Ordinary Level (G.C.E. O/L) and Advanced Level (G.C.E. A/L) examinations in Sri Lanka face severe systemic challenges. The country faces a shortage of more than 30,000 qualified teachers, causing student-to-teacher ratios to exceed 50:1, with the problem disproportionately affecting rural provinces [1]. These conditions perpetuate a stark divide in educational quality between urban and rural learners. Statistically, only 63.3% of O/L candidates qualify for A/L studies, blocking over 130,000 students annually from tertiary education pathways, while a mere 15% of A/L candidates secure university admission [2].

Several digital platforms have attempted to bridge this divide. The government-sponsored e-Thaksalawa portal [3] and private initiatives such as Dammika Perera (DP) Education offer pre-recorded video lectures, downloadable notes, and fixed question banks but lack the adaptive, interactive dimension necessary for effective personalized tutoring. General purpose Large Language Models (LLMs), while increasingly capable in Sinhala, are not aligned with the

Sri Lankan national curriculum and tend to generate generic responses inappropriate for exam preparation [4].

These observations point to a research gap that, to the best of our knowledge, has not been addressed in the literature: The absence of an Artificial Intelligence (AI) based educational system that (a) operates natively in the Sinhala medium, (b) strictly adheres to national syllabus content, (c) dynamically adapts to each student's evolving knowledge state, and (d) provides verifiable, curriculum-grounded responses rather than unchecked generative output, are the observations that point to the research gap.

This paper addresses the identified gap by proposing an AI-based Sinhala teaching assistant designed for G.C.E. O/L and A/L students. The system brings together four core components: a fine-tuned Sinhala LLM (SinLlama-7B) [5] for content generation, a Hybrid- Bayesian Knowledge Tracing (BKT) personalization engine that merges Personalized Clustered BKT with Long Short-Term Memory (LSTM) based temporal modelling for mastery estimation, a Retrieval Augmented Generation (RAG) pipeline that anchored in a ChromaDB vector database, and a LangGraph based multi agent layer that coordinates specialized teaching agents.

The work is guided by four research questions investigating: (i) the effectiveness of different LLM architectures in achieving high semantic fidelity for curriculum-aligned Sinhala content generation after fine-tuning on national syllabus data, (ii) whether a hybrid ensemble combining Personalized Clustered BKT and LSTM-based temporal modeling improves student mastery estimation compared to conventional BKT approaches, (iii) the extent to which a RAG pipeline grounded in vectorized syllabus content can constrain model outputs within curriculum boundaries, and (iv) the ability of a multi-agent architecture to effectively orchestrate personalized and adaptive learning workflows for secondary-level education. Finally, a deployed web and mobile prototype is presented to demonstrate practical applicability.

The remainder of this paper is organized as follows. Section II reviews related work across intelligent tutoring systems, adaptive learning, LLMs in education, RAG, and multi-agent architectures. Section III details the system methodology, Section IV reports experimental results. Section V discusses findings and limitations, and Section VI concludes with directions for future work.

II. RELATED WORK

This section reviews five areas of related research: intelligent tutoring systems, adaptive learning and student modelling, generative AI and LLMs in education, retrieval-augmented generation for education, and multi-agent tutoring architectures.

A. Intelligent Tutoring Systems

Intelligent Tutoring Systems (ITS) employ machine learning, educational data mining, and adaptive assessment to personalize instruction [6]. Xu et al. [7] demonstrated that AI-driven adaptive learning frameworks significantly improve engagement, assessment scores, and knowledge retention compared to non-adaptive systems. Liu et al. [8] reported that supervised Machine Learning (ML) combined with Natural Language Processing (NLP) and Bayesian Knowledge Tracing reduces cognitive load and increases learner satisfaction. Alshahrani et al. [9] further emphasized that deep learning models can enhance personalized adaptive learning by capturing complex learner behavior patterns. However, most ITS implementations target English-language learners via text-only interaction, leaving low-resource language contexts underserved [10].

B. Adaptive Learning and Student Modeling

BKT introduced by Corbett and Anderson [11], remains the most widely adopted student modelling framework. BKT estimates the probability of knowledge mastery through a Hidden Markov Model parameterized by four parameters: Prior knowledge $P(L_0)$, learning rate $P(T)$, guess rate $P(G)$, and slip rate $P(S)$. However, standard BKT assumes uniform learner priors and fixed parameters, limiting its capacity for personalized student modelling.

Several extensions address these limitations. Pardos and Heffernan [12] introduced personalized BKT parameters per student, demonstrating significant accuracy improvements on the ASSISTments dataset. Nedungadi and Remya [13] proposed Personalized Clustered BKT (PC-BKT), combining K-Means based student clustering with empirically estimated BKT parameters to improve cold-start predictions. Deep Knowledge Tracing (DKT) by Piech et al. [14] reformulated knowledge tracing as a sequence prediction problem using recurrent neural networks, achieving strong performance but sacrificing the interpretability of probabilistic models. Pandey and Karypis [15] proposed Self-Attentive Knowledge Tracing (SAKT), applying transformer based attention mechanisms to student interaction sequences. Minn [16] combined BKT with LSTM networks (BKT-LSTM), capturing temporal learning dynamics beyond the Markov assumption.

C. Generative AI and LLMs in Education

Recent systematic reviews confirm that LLMs such as GPT, BERT, and T5 offer significant potential for personalized learning, adaptive feedback, and intelligent tutoring [17]. Kasneci et al. [18] highlighted both the promise and risks of ChatGPT in education, noting its tendency to hallucinate and its lack of curriculum alignment.

For low resource languages such as Sinhala, the availability of pre trained language models has improved with the development of SinLlama [5], a 7-billion-parameter model specifically pre trained on Sinhala text corpora. However, no prior work has fine-tuned SinLlama or any Sinhala LLM on national curriculum content for structured examination preparation. These limitations motivate the need for curriculum-grounded, language-specific fine-tuning as pursued in this work.

D. Retrieval-Augmented Generation for Education

RAG combines information retrieval with generative models to reduce hallucinations and improve factual grounding. The foundational RAG framework was introduced by Lewis et al. [19] demonstrating that conditioning generation on retrieved documents substantially improves knowledge-intensive NLP tasks. In the educational domain, Swacha and Gracel [20] demonstrated that RAG-based educational chatbots produce syllabus-aligned answers that enhance student trust. Lian [21] showed that RAG based learning assistants improve comprehension versus purely generative models. Li et al. [22] systematically reviewed RAG techniques for education, noting their suitability for curriculum based adaptive learning. However, existing RAG based educational systems focus almost entirely on English and higher education contexts. Applying RAG to Sinhala medium secondary school education where textbooks must be digitized, segmented, and embedded in a low resource language pipeline remains unexplored.

E. Multi-Agent Systems for Tutoring

Multi-agent architectures distribute teaching tasks among specialized agents, improving system scalability and pedagogical flexibility. David and Ghosh [23] introduced IntelliCode, a StateGraph based orchestration architecture with centralized learner memory that enables long-term student tracking across multiple tutoring sessions. Glossner et al. [24] proposed a multi agent mathematics learning framework featuring dedicated teacher, tutor, solution verifier, level adapter, and motivator agents powered by GPT-4o, demonstrating that role decomposition yields structured, curriculum consistent content delivery. The LangGraph framework [25] extends LangChain with stateful, graph-based orchestration of multi agent workflows, built-in checkpointing, and conditional routing making it well-suited for adaptive educational applications that require persistent student state management.

F. Summary of Research Gaps

A review of existing systems reveals several significant gaps in Sinhala-medium intelligent tutoring systems. To the best of our knowledge, no existing solution system simultaneously integrates Sinhala medium instruction, national curriculum alignment, adaptive student modelling, RAG-based content grounding, and multi agent orchestration within a unified educational framework.

III. METHODOLOGY

A. System Architecture Overview

The proposed system follows a modular architecture comprising five integrated layers, as illustrated in Fig. 1. The functionality of each layer is described below:

1. **User Interface Module** - Consisting of student, teacher, and parent dashboards served on web and mobile platforms.
2. Multi-Agent AI Layer - Consisting of task-specific agents orchestrated via LangGraph for adaptive workflow management.
3. RAG-based Knowledge Engine - Retrieval pipeline that grounds all generated content in vectorized Sinhala textbooks.
4. Hybrid-BKT Student Modelling and Adaptive Learning Engine - A personalization engine for mastery estimation and adaptive content delivery.

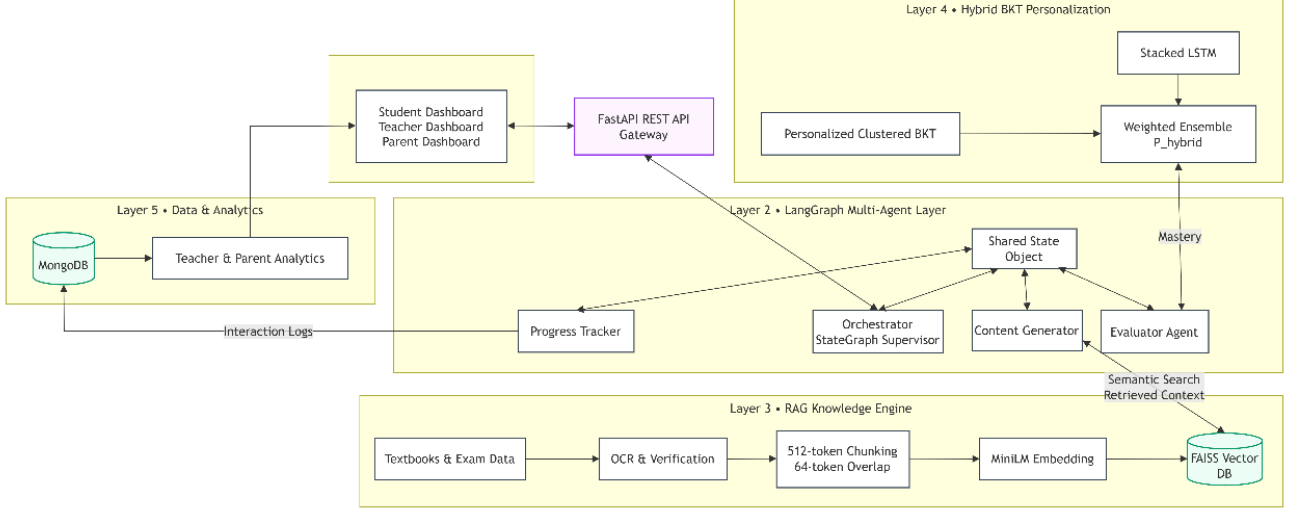


Fig 1: System Architecture Diagram

5. Teacher Analytics Dashboard - Monitoring interface for educators to track student progress and identify learning gaps.

Each layer communicates through well-defined REST API endpoints implemented with FastAPI, ensuring horizontal scalability and component modularity. Fig. 1 illustrates the data flow between layers.

B. LLM Fine-Tuning for Sinhala Education

To identify the most suitable model for Sinhala content generation, five Sinhala-supporting LLMs were evaluated. SinLlama-7B, Qwen-14B, Llama-3.1-8B, DeepSeek-14B, and Mistral-7B. All five LLMs were fine-tuned on the Buddhism Questions and Answers dataset using Low-Rank Adaptation (LoRA) [26] to enable training on consumer-grade hardware.

Fine-tuning was executed for 3 epochs with a maximum sequence length of 2,048 tokens, utilizing a LoRA rank (r) of 16, alpha (α) of 32, a LoRA dropout of 0.05, and the AdamW optimizer (learning rate 2×10^{-4} , weight decay 0.01, warmup ratio 0.03) with a batch size of 4 (gradient accumulation: 4).

All models were fine-tuned on this dataset and evaluated using ROUGE scores, BERTScore, token generation speed, and an LLM-as-a-Judge protocol, where a larger-parameter LLM acted as the evaluator and assigned scores out of 20 based on Accuracy, Completeness, Clarity, and Depth. The model demonstrating the best overall performance in the evaluation phase was subsequently selected for integration into the proposed system.

C. Retrieval-Augmented Generation Pipeline

The RAG pipeline ensures all generated content remains grounded in verified curriculum resources, thereby mitigating the hallucination risks commonly associated with standalone generative language models.

(i) The knowledge base was constructed from Sinhala-medium educational resources, including official textbooks, teacher guidance manuals, and past examination papers. Source documents were digitized using Optical Character Recognition (OCR) and subsequently verified to ensure textual accuracy. To preserve semantic continuity while

enabling efficient retrieval, documents were segmented into overlapping chunks of 512 tokens with an overlap of 64 tokens. During document ingestion, textbook illustrations and diagrams are preserved as specialized tokens (e.g., [IMAGE: image_name.png]).

(ii) Each document chunk was encoded using the `intfloat/multilingual-e5-base` sentence transformer model, producing 768-dimensional dense vector representations. The resulting embeddings were indexed using ChromaDB.

(iii) During inference, student queries and assessment answers were embedded through the same model as that of the documents and matched to the indexed document repository. The top five document chunks similar to the vector were retrieved and used in the generation prompt. Mastery awareness-based prompting was adopted in the process, combining the learner's level of mastery. This approach enables the framework to facilitate generation of pedagogically appropriate responses by adding context to generation with the curriculum-relevant information.

The backend FastAPI server mounts a static images directory, enabling the Next.js frontend to dynamically render these illustrations inline with the retrieved text, maintaining a multimodal learning experience.

D. Hybrid-BKT Personalization Engine

The personalization engine uses a weighted ensemble of two complementary models: Personalized Clustered Bayesian Knowledge Tracing (PC-BKT) and a BKT-LSTM temporal predictor.

(a) *PC-BKT*: Standard BKT is a state-space model that tracks a student's latent knowledge state over time. It is governed by four global parameters:

$$\begin{aligned} P(L_0) &- \text{Initial mastery probability} \\ P(T) &- \text{Learning transition probability} \\ P(G) &- \text{Guess probability} \\ P(S) &- \text{Slip probability} \end{aligned}$$

When a student interacts with the system at timestep t , the correctness indicator $C_t \in \{0, 1\}$ is observed, and the probability of mastery is updated dynamically.

Correct response Update:

$$P(L_t | C_t = 1) = \frac{P(L_{t-1})(1 - P(S))}{P(L_{t-1})(1 - P(S)) + (1 - P(L_{t-1}))P(G)}$$

Incorrect response Update:

$$P(L_t | C_t = 0) = \frac{P(L_{t-1})P(S)}{P(L_{t-1})P(S) + (1 - P(L_{t-1}))(1 - P(G))}$$

Learning Transition Update:

$$P(L_{t+1}) = P(L_t | C_t) + (1 - P(L_t | C_t))P(T)$$

A limitation of standard BKT is that the parameters $P(L_0)$, $P(T)$, $P(G)$, and $P(S)$ are shared across all learners. PC-BKT [13] extends standard BKT with three innovations.

First, personalized initial mastery priors, $P(L_0)$ are computed per student knowledge components (KC) pair via Correct First Attempt (CFA) rates rather than a fixed default. knowledge components (KC) are derived from the competency levels specified in the national teacher guidance manual. These competency levels have been defined with respect to specific knowledge domains within each subject, allowing a thorough assessment to be performed for the entire curriculum.

Second, students are clustered into $K = 3$ ability groups (High, Medium, Low) by applying K-Means++ clustering on a capability matrix.

$$B_{ij} \in \mathbb{R}^{(N \times KC)}$$

B_{ij} denotes the proportion of correct responses by student i on knowledge component j .

N : Number of Students

KC : Number of Knowledge components

Third, guess $P(G)$, slip $P(S)$, and transition $P(T)$ probabilities are estimated empirically from annotated interaction sequences via maximum likelihood estimation rather than manual presets.

(b) BKT-LSTM: The BKT-LSTM component [16] reformulates mastery prediction as a temporal sequence forecasting problem. At each interaction timestep t , a feature vector is constructed:

$$f(t) = [P(L)_{BKT}, d_{KC}, c_t, a_t]$$

where $P(L)_{BKT}$ is the current standard BKT mastery probability, d_{KC} is the normalized difficulty of the knowledge component, $c_t \in \{0, 1\}$ is the correctness indicator, and a_t is the cumulative attempt count.

The BKT-LSTM model employs a two-layer stacked LSTM (128 hidden units per layer) to capture temporal learning patterns, including forgetting effects and cross-knowledge component transfer, which are not represented by standard BKT.

(c) Hybrid-BKT: The hybrid student mastery estimate at timestep t , denoted as $Hybrid(t)$, is calculated as a weighted ensemble of the PC-BKT and BKT-LSTM predictions:

$$P_{\{hybrid\}}(t) = \alpha P_{\{PC-BKT\}}(t) + (1 - \alpha)P_{\{LSTM\}}(t)$$

Here, $\alpha \in [0, 1]$ represents the ensemble weight parameter that determines the relative contribution of the PC-BKT and BKT-LSTM models. The optimal value of α was determined to be 0.665 using a two-stage grid search designed to minimize the Root Mean Squared Error (RMSE) on a held-

out validation fold. The weight $\alpha = 0.665$ was obtained by first running a coarse grid search over: $\alpha \in \{0.0, 0.05, \dots, 1.0\}$ which identified the local minimum in the range $[0.60, 0.70]$.

E. Multi-Agent Orchestration

The adaptive learning workflow is orchestrated by a **LangGraph StateGraph** supervisor coordinating four dedicated agents.

- The Orchestrator Node: determines learner state and routes data based on quiz type, scoring completion, content availability, and mastery progression.
- The Evaluator Agent: classifies students as Beginner ($< 50\%$), Intermediate ($\geq 50\%$), or Advanced ($\geq 80\%$) based on quiz scores and updates BKT states.
- Content Generator Agent: uses the RAG pipeline to produce mastery-aware personalized lessons.
- Progress Tracker Agent: maintains persistent interaction logs, mastery histories, and engagement analytics in MongoDB.

All agents share a common state object managed by LangGraph, ensuring consistent learner context across transitions.

F. Adaptive Assessment Cycle

The proposed framework facilitates learning through four phases.

Phase 1 (Pre-Assessment):

Students complete a pre-quiz on a target knowledge component; questions are generated dynamically via an LLM-based pipeline aligned to Bloom's taxonomy (Remember, Understand, Apply), with one question per level drawn from curriculum metadata.

Phase 2 (Adaptive Content Generation):

The Evaluator Agent classifies mastery via $P_{\{hybrid\}}(t)$ and assigns one of the three tiers. Beginner ($P_{\{hybrid\}}(t) < 0.60$): foundational content with definitions, worked examples, and guided practice. Intermediate ($0.60 \leq P_{\{hybrid\}}(t) < 0.85$): standard lesson with moderate scaffolding and application exercises. Advanced ($P_{\{hybrid\}}(t) \geq 0.85$): enrichment content with higher-order analysis and past-paper questions. The Evaluator Agent classifies mastery level and the Content Generator Agent produces a personalized lesson.

Phase 3 (Post-Assessment):

A post-quiz of equivalent difficulty measures updated mastery $P_{\{hybrid\}}(t+1)$. Students who meet the advancement threshold (≥ 0.85) progress to the next KC; those in the Intermediate range repeat the KC with alternative content; those below 0.60 receive remedial material with simplified explanations

Phase 4 (Continuous Adaptation):

Student interactions continuously update the capability matrix, while periodic retraining refines learner ability clusters to improve personalization over time..

IV. RESULTS

A. LLM Evaluation

Five Sinhala-supporting LLMs were fine-tuned on the Buddhism curriculum question and answering (QA) dataset

comprising 3,547 QA pairs. The dataset was synthetically generated using AI based on the national subject content and curriculum materials provided for the Buddhism syllabus. The models were evaluated using ROUGE scores, BERTScore, token generation throughput (measured as the number of tokens generated per second), and an LLM-as-a-Judge evaluation protocol, where a separate LLM was employed to assess the quality, correctness, relevance, and pedagogical appropriateness of generated responses according to predefined evaluation criteria.

The evaluation was conducted on a held-out test set of 532 QA pairs (15% of the dataset), stratified by topic to ensure coverage across all KCs. The results have been obtained by calculating the average performance across three runs of fine tuning with random seeds.

Table I presents ROUGE scores for all the tested five models. SinLlama-7B demonstrates the best overall performance across all ROUGE metrics, achieving ROUGE-1 of 0.7714, ROUGE-2 of 0.6418, and ROUGE-L of 0.6111. It significantly outperforms the next-best baseline, Mistral, which records a ROUGE-1 score of 0.4925. Table II presents the BERTScore results, where SinLlama-7B achieves the highest score of 0.9424, consistently outperforming Llama-3.1, DeepSeek-14B, Mistral-7B, and Qwen-14B, with scores of 0.8799, 0.8659, 0.8555, and 0.8022, respectively. In the LLM-as-a-Judge evaluation (Table II), SinLlama-7B achieved the highest overall score of 15.88/20, followed by Mistral-7B with 14.97/20. DeepSeek demonstrated the fastest token generation (444.9 tokens/sec) but lower semantic accuracy, making it suitable only for latency-critical deployments where accuracy is secondary.

TABLE I. ROUGE SCORES OF EVALUATED LLMs

Model	ROUGE-1	ROUGE-2	ROUGE-L
SinLlama-7B	0.7714*	0.6418*	0.6111*
Qwen-14B	0.3310	0.1678	0.1059
Llama-3.1-8B	0.3983	0.1949	0.1418
DeepSeek-14B	0.4485	0.2304	0.1845
Mistral-7B	0.4925	0.2280	0.1594

* STATISTICALLY SIGNIFICANT VS. ALL OTHER MODELS (WILCOXON SIGNED-RANK, $P < 0.001$).

TABLE II. BERTSCORE AND LLM-AS-A-JUDGE EVALUATION

Model	BERT Score	Accuracy	Clarity	Depth	Total/20
SinLlama-7B	0.9424	4.78	4.50	2.38	15.88
Mistral-7B	0.8555	4.50	4.34	2.53	14.97
Llama-3.1-8B	0.8799	4.28	4.34	2.25	13.59
DeepSeek-14B	0.8659	4.22	4.36	2.09	13.58
Qwen-14B	0.8022	2.97	4.34	1.84	11.75

DeepSeek-14B demonstrated the highest throughput of 444.9 tokens/sec compared to SinLlama-7B at 127.3 tokens/sec; however, its lower semantic accuracy makes it suitable only for latency-critical deployments where content precision is secondary. On the strength of these results,

SinLlama was selected as the production model, for the proposed work.

B. Hybrid-BKT Personalization Evaluation

The Hybrid-BKT engine was evaluated on the synthetic dataset of 6,532 simulated student interactions across 47 KC derived from the competency levels. It should be noted that real classroom data was unavailable at this stage; implications of relying on synthetic data are discussed.

Table III compares performance across three BKT variants using 5-fold cross-validation stratified by simulated student identity. Hybrid-BKT achieved the highest classification accuracy of 56.2% and the lowest RMSE of 0.49.

TABLE III. PERFORMANCE COMPARISON OF BKT VARIANTS

Metric	Standard BKT	BKT-LSTM	Hybrid-BKT
AUC	0.60	0.59	0.60
Accuracy	48.1%	54.7%	56.2%
RMSE	0.51	0.53	0.49

Although BKT-LSTM achieved slightly lower performance than Standard BKT (AUC: 0.59 vs. 0.60; RMSE: 0.53 vs. 0.51), this is likely attributable to the limited size and homogeneity of the synthetic dataset, which restricted the model's ability to learn meaningful temporal patterns. Nevertheless, the LSTM component contributed positively within the Hybrid-BKT ensemble by capturing complex learning dynamics, including mastery regression and non-monotonic learning behaviors.

Table IV presents a sensitivity analysis of the ensemble weight α . The weight was optimized via grid search over $\alpha \in \{0.0, 0.05, \dots, 1.0\}$ to minimize validation-fold RMSE. The optimal value $\alpha = 0.65$ yielded the best accuracy and lowest RMSE.

TABLE IV. ENSEMBLE WEIGHT SENSITIVITY ANALYSIS

α (PC-BKT weight)	Accuracy (%)	RMSE	AUC
0.00 (LSTM only)	54.7	0.53	0.59
0.35	55.1	0.51	0.59
0.50	55.8	0.50	0.60
0.65 (selected)	56.2	0.49	0.60
0.80	55.4	0.50	0.60

V. DISCUSSION

The results confirm that this work makes several novel contributions to AI-based educational systems, particularly in enabling personalized learning and Sinhala-language-based educational support.

First, the Hybrid-BKT model demonstrates that combining probabilistic student modeling with temporal LSTM prediction yields higher accuracy compared to either technique in isolation. The weighted ensemble (0.65 PC-BKT + 0.35 LSTM) preserves the pedagogical interpretability of BKT while capturing temporal learning dynamics that rule-based models miss. To our knowledge, this combination has not previously been applied in any Sinhala-medium educational system.

Second, SinLlama's dominant performance across all evaluation metrics validates the strategy of selecting a

language-specific model over general-purpose multilingual LLMs for low-resource language educational applications. SinLlama's ROUGE-1 score of 0.7714 is approximately 56% higher than the next-best general model, Mistral-7B. Underscoring the critical importance of language-specific fine-tuning for curriculum-aligned content generation.

Third, RAG pipeline addresses a fundamental limitation of pure generative models in educational contexts: hallucination. By grounding all generated content in vectorized Sinhala textbooks, the system constrains responses to curriculum boundaries and provides verifiable sources. While initial testing was focused on Grade 11 Buddhism, the system's curriculum engine has been extended capability to generalize to other subjects as well.

Several limitations should be acknowledged. The Hybrid-BKT accuracy of 56.2%, while superior to baselines, indicates room for improvement with larger, real-world interaction datasets. **Text-to-Speech (TTS), Automatic Speech Recognition (ASR)** and Avatar-based interaction capabilities remain basic. Real-classroom validation with genuine students and teachers remains as future work.

VI. CONCLUSION

This paper presents an AI-based Sinhala teaching assistant for personalized G.C.E. O/L and A/L learning. The system combines a fine-tuned Sinhala LLM (SinLlama-7B), a Hybrid-BKT personalization engine, a curriculum-grounded RAG pipeline, and a LangGraph multi-agent orchestration architecture. SinLlama-7B achieved the highest semantic performance among five evaluated LLMs ROUGE-1: 0.7714, BERTScore: 0.9424, Judge score: 15.88/20). The Hybrid-BKT model outperformed Standard BKT and BKT-LSTM baselines with 56.2% accuracy and 0.49 RMSE, with a cluster-based cold-start strategy reducing initial prediction error by 12%. The fully functional web and mobile prototypes include student learning interfaces, teacher analytics, and parent monitoring.

Future work will focus on expanding the knowledge base to all O/L and A/L subjects, improving personalization accuracy with large-scale real classroom data, enhancing Sinhala TTS and ASR components. Federated learning will be explored to enable personalized recommendations while preserving student data privacy. Longitudinal classroom trials with real students, teachers, and parents will provide the empirical validation needed to establish educational effectiveness.

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